

High-Dimensional Instrumental Variable Quantile Regression under Weak Instruments: A Monte Carlo Study

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Supervisor:
JProf. Claudia Noack Ph.D.

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Olimjon Umurzokov

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Abstract

This thesis studies the finite-sample consequences of weakening excluded-instrument relevance in high-dimensional instrumental variable quantile regression (IVQR). Building on a preserved DML-IVQR Monte Carlo design, instrument strength is varied through $\kappa \in \{1, 0.50, 0.25, 0.10\}$ while holding the marginal treatment distribution, endogeneity channel, structural outcome equation, and control dimension fixed. Oracle-GMM, Full-GMM, and DML-IVQR-BC are compared across five quantiles and $n \in \{500, 1000\}$. As κ falls, MAE and RMSE rise, accepted sets expand, and power against false treatment-effect values declines for all estimators, including the oracle benchmark. Coverage is also imperfectly calibrated in some finite-sample cells, even under strong relevance. DML often improves on Full-GMM but does not uniformly match Oracle-GMM. The results show that orthogonalization and regularized nuisance estimation can reduce high-dimensional estimation costs, but they cannot replace identifying information lost when instruments become weakly relevant.

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1 Introduction

Average treatment effects can conceal economically important heterogeneity. A policy, program, or endogenous choice may have a modest effect around the center of an outcome distribution while having substantially different effects for observations near the lower or upper tail. Quantile treatment effects provide a way to describe this heterogeneity by asking how the structural effect varies across different quantiles of the potential-outcome distribution. When treatment is endogenous, however, conventional quantile regression does not recover the structural effect because treatment choices may be related to unobserved determinants of the outcome. Instrumental variable quantile regression (IVQR) addresses this problem by combining excluded instruments with restrictions on the structural ranks of potential outcomes (Chernozhukov and Hansen 2005). The resulting framework allows the treatment effect itself to vary across quantiles while maintaining an explicit treatment-endogeneity problem.

Two additional complications arise in modern applications. First, empirical specifications may contain many controls, transformations, and interactions. Their coefficients are nuisance parameters rather than the primary object of interest, but estimating a large nuisance model without structure can be unstable and may contaminate inference on the treatment effect. Regularization provides one way to estimate such high-dimensional components, while Neyman-orthogonal scores reduce the first-order sensitivity of the target moment to sufficiently small nuisance-estimation errors (Chernozhukov et al. 2018). Chen et al. (2021) combine these ideas with IVQR and develop a DML-IVQR procedure that accommodates high-dimensional controls while conducting candidate-specific inference on the structural quantile effect.

Second, regularization does not solve weak identification. Even a valid excluded instrument may contain little information about the endogenous treatment. When instrument-induced treatment variation is small, different candidate treatment-effect values can generate similar population moments, making them difficult to distinguish statistically. This problem is particularly important in IVQR because the relevant identifying restrictions are quantile specific and nonlinear. Conventional first-stage measures can describe the strength of the instrument–treatment relationship, but they do not by themselves characterize the amount of information available for identifying a structural quantile effect. The broader weak-instrument literature shows why conventional estimate-centered approximations can become unreliable when identifying information is weak, and why candidate-specific procedures can remain useful even when point identification is imprecise (Staiger and Stock 1997; Stock and Wright 2000; Andrews et al. 2019). In IVQR, this distinction is reflected in weak-identification-robust

test inversion: a procedure may remain asymptotically valid while producing broad sets of accepted candidate values and little power against economically relevant false alternatives.

These two problems—high-dimensional nuisance estimation and weak instrument relevance—are conceptually different. Neyman orthogonality is designed to protect the target moment against first-order errors in nuisance estimation. It does not generate additional treatment variation or strengthen the mapping between instruments and the structural parameter. Conversely, a weak-identification-robust testing procedure is designed to avoid relying on strong local identification, but robustness of the reference distribution does not imply that the resulting inference will be precise or informative in finite samples. Distinguishing these mechanisms is important for evaluating DML-IVQR. If performance deteriorates as instruments weaken, the deterioration may reflect a loss of structural information even when nuisance estimation is handled well.

This distinction motivates the empirical question studied in this thesis. Existing Monte Carlo evidence for DML-IVQR evaluates regularized nuisance estimation in a high-dimensional design with a fixed level of instrument relevance. It therefore does not systematically show how the same estimators and inference procedures behave when excluded-instrument relevance is progressively weakened while the rest of the structural environment is held fixed. The central research question is: How does weakening instrument strength affect the finite-sample performance and informativeness of weak-identification-robust inference in DML-IVQR, and how do these effects vary across quantiles?

The thesis answers this question through a controlled extension of the Monte Carlo design in Chen et al. (2021). A single parameter,

$$\kappa \in \{1.00, 0.50, 0.25, 0.10\},$$

controls the loading of the excluded-instrument primitives in the latent treatment equation. Smaller values of κ therefore reduce excluded-instrument relevance within the simulation. The construction simultaneously adds an independent component to the treatment index so that its marginal variance remains unchanged as κ varies. The endogeneity channel, structural outcome equation, true quantile treatment effects, control dimension, and observed instrument construction are also preserved. This design is important because it makes the four relevance settings directly comparable: the experiment changes the contribution of the excluded instruments to treatment while avoiding a simultaneous change in the marginal treatment distribution or the strength of the outcome–treatment endogeneity channel. The parameter κ is there-

fore a controlled relevance parameter for this Monte Carlo design, not a universal scalar measure of IVQR identification strength.

Three estimators are compared on identical simulated samples. Oracle-GMM is an infeasible benchmark that knows the designated relevant control set. Full-GMM uses the full high-dimensional control vector without regularization. DML-IVQR-BC uses regularized nuisance estimation together with the preserved Belloni–Chernozhukov pivotal penalty implementation. The comparison separates the common effect of declining instrument information from estimator-specific consequences of nuisance estimation, although differences between Full-GMM and DML-IVQR-BC cannot be attributed to regularization alone because their residualization implementations also differ. Performance is evaluated across five quantiles, $\tau \in \{0.10, 0.25, 0.50, 0.75, 0.90\}$, and two sample sizes, $n \in \{500, 1000\}$.

The evaluation deliberately separates several dimensions of finite-sample performance. Bias, mean absolute error, and root mean squared error summarize point-estimation accuracy. Coverage evaluates whether the true structural quantile effect is retained by the candidate-specific test. Power measures the ability to reject false treatment-effect values. Inferential informativeness is summarized by the grid-based accepted-set measure within the prespecified primary computational domain $A_0 = [-1, 3]$. These quantities answer different questions. High coverage is not necessarily informative if the procedure also accepts a large fraction of false candidates, while narrow accepted sets are not useful if the true value is rejected too frequently. Examining accuracy, coverage, power, and accepted-set informativeness separately therefore gives a more complete picture of how inference changes as excluded-instrument relevance weakens.

The Monte Carlo evidence shows a consistent deterioration in informativeness as relevance declines. Point-estimation error rises substantially for Oracle-GMM, Full-GMM, and DML-IVQR-BC; accepted sets expand; and power against false candidate values falls. The deterioration of the oracle benchmark is particularly informative because it shows that the loss is not generated only by high-dimensional nuisance estimation. At the same time, the magnitude of the deterioration varies across quantiles and estimators. Coverage is also not uniformly close to its nominal level in finite samples, including in some strong-relevance cells, showing that weak-identification robustness of the asymptotic procedure does not imply exact finite-sample calibration. DML-IVQR-BC often performs better than the unregularized high-dimensional benchmark, but it does not dominate in every design cell and cannot compensate for the loss of identifying information when excluded instruments become only weakly relevant. The main contribution of the thesis is therefore to document, within a controlled high-dimensional IVQR experiment, how nuisance robustness, instrument relevance, inferential informativeness, and

finite-sample calibration interact as the information supplied by the excluded instruments is systematically reduced.

2 Literature Review

Quantile regression describes how covariates are related to different parts of a conditional outcome distribution rather than only its mean (Koenker and Bassett 1978). This distinction matters when treatment effects are heterogeneous: an intervention may shift lower and upper quantiles by different amounts even when its average effect is modest. A structural quantile treatment effect compares the same quantile of potential-outcome distributions under alternative treatment states. It is generally a distributional effect, not an individual-level effect, because the observation occupying a given rank can differ across counterfactual states. Endogeneity creates a separate problem. When realized treatment depends on latent determinants of the outcome, the conditional distribution among observations selecting a treatment value need not equal the potential-outcome distribution obtained by fixing treatment at that value. Ordinary quantile regression of the outcome on treatment and controls then estimates an observed conditional association and does not generally recover a structural quantile effect. The IVQR framework of Chernozhukov and Hansen (2005) uses excluded instruments to connect endogenous treatment variation to structural outcome quantiles.

Structural IVQR represents potential outcomes through a quantile function and a latent structural rank. For a treatment state d , the structural rank $U_d \in (0, 1)$ is the latent quantile index that locates an individual within the conditional distribution of the potential outcome Y_d given observed controls. Because the structural quantile function is increasing in this rank, a larger value of U_d corresponds to a higher position in the potential-outcome distribution. The rank may be interpreted as summarizing unobserved heterogeneity that determines where an observationally similar individual lies in that distribution. Instrument exogeneity restricts the relationship between the instruments and these structural ranks conditional on controls, while treatment may still depend on latent variables related to the outcome. Rank invariance assumes that an individual's rank is unchanged across treatment states; the weaker rank-similarity condition allows ranks to change but requires their conditional distributions to agree across those states. Under the maintained selection, exogeneity, and rank conditions, the probability that the observed outcome lies below the true structural quantile, conditional on controls and instruments, equals the quantile index (Chernozhukov and Hansen 2005; Chernozhukov et al. 2017).

This conditional restriction links the potential-outcome model to estimable instrument moments. Validity permits the instrument to enter those moments; relevance determines whether instrument-induced treatment variation separates competing structural coefficients. Moment validity alone does not establish point identification. A unique solution additionally requires the population mapping from candidate structural quantiles to the restrictions to distinguish the truth from alternatives. That mapping is quantile specific and density weighted, so a mean first-stage statistic is informative about relevance but does not completely characterize IVQR identification at a particular quantile. This distinction is essential empirically: an instrument can satisfy the maintained exogeneity condition yet generate too little treatment variation to locate a structural quantile effect precisely.

When identifying information is strong, the population moment moves sufficiently as the candidate treatment coefficient moves away from its true value, supporting regular point-estimator approximations. With weak information, the true value may remain unique while nearby candidates generate almost indistinguishable moments. Partial identification instead permits more than one population solution, and failed identification leaves some candidates or directions indistinguishable. Chernozhukov and Hansen (2008) develop inference that remains asymptotically valid without imposing the strong-identification conditions needed for conventional estimate-centered inference. Their approach fixes a candidate value, estimates or profiles the nuisance parameters under that candidate, and tests whether the resulting instrument-based quantile moments are compatible with zero. Inverting these candidate-specific tests retains every non-rejected value and can therefore produce asymmetric, disconnected, or very wide confidence regions. Weak-identification robustness concerns the null reference distribution; it does not guarantee that false candidates will be rejected. Informativeness and finite-sample calibration remain distinct: a robust confidence region can honestly reveal little about the parameter, and an asymptotic test can still reject too often or too rarely in a finite sample. The exact conditional procedure of Chernozhukov et al. (2009) uses a different simulated reference distribution and is not the method implemented here.

The IVQR inference used in this thesis also belongs to a broader literature on inference under weak instrumental-variable identification. Anderson and Rubin (1949) provide the classical foundation for testing structural parameter values through restrictions that remain meaningful without relying on a precisely estimated first stage. Staiger and Stock (1997) formalize weak-instrument asymptotics in linear instrumental-variables models and show why conventional estimate-centered approximations can become unreliable when instruments are only weakly correlated with endogenous regressors. Stock and Wright (2000) extend weak-identification analysis to generalized method-of-moments settings and develop confidence pro-

cedures that remain valid when parameters are weakly identified. Alternative robust procedures include the pivotal testing approach of Kleibergen (2002) and the conditional likelihood-ratio method of Moreira (2003). Andrews et al. (2019) provide a modern review of this literature and its implications for applied work. These results motivate the general distinction between conventional estimate-centered inference and candidate-specific procedures that remain valid under weak identification. They do not, however, provide a complete characterization of identification in IVQR. In particular, linear-IV first-stage diagnostics and threshold rules do not directly measure quantile-specific IVQR identification, whose population moments are nonlinear and depend on the distribution of the outcome around the quantile of interest. The thesis therefore uses first-stage statistics only to describe the designed ordering of excluded-instrument relevance, while the IVQR-specific weak-identification-robust inference follows Chernozhukov and Hansen (2008).

High-dimensional controls create a second challenge because estimating the outcome and instrument nuisances may require many coefficients relative to the sample size. Regularization exploits approximate sparsity: a comparatively small set of variables captures most predictive content, while the remaining coefficients are zero or sufficiently small. Shrinkage and selection can nevertheless contaminate a target moment. Double machine learning addresses this problem with a Neyman-orthogonal score whose population expectation has zero first derivative with respect to local nuisance perturbations at the truth (Chernozhukov et al. 2018). Orthogonality therefore removes the leading sensitivity to sufficiently small nuisance-estimation errors, not the errors themselves; the learners must still satisfy suitable convergence and regularity conditions. Sample splitting estimates nuisances on one subsample and evaluates the score on another, while cross-fitting rotates these roles across folds. These concepts should not be conflated: orthogonality is a property of the score, whereas cross-fitting organizes data use and limits overfitting dependence. Nor does DML strengthen an instrument. It addresses high-dimensional nuisance estimation, while candidate-specific robust inference addresses irregular inference under limited identification.

Chen et al. (2021) join these two strands. For each candidate, they profile a high-dimensional quantile regression, construct a density-weighted instrument residual, form a Neyman-orthogonal IVQR score, estimate the nuisances with regularization, and evaluate candidates by GMM and weak-identification-robust test inversion. Although they discuss cross-fitting as part of the broader DML framework, the executable DML-IVQR algorithm preserved in this thesis estimates nuisances and evaluates the score on the same sample. Their principal Monte Carlo design holds instrument relevance fixed while comparing Oracle-GMM, Full-GMM, and DML-IVQR across several quantiles. The remaining empirical question is how

point-estimation error, confidence-set informativeness, power, and finite-sample calibration respond when excluded-instrument relevance is systematically weakened within that otherwise preserved design. This thesis supplies that controlled stress test; it does not propose a new structural model, score, estimator, identification theorem, or finite-sample correction. Jin and Sun (2025) study a related nonparametric high-dimensional IVQR problem for unconditional quantile treatment effects with automatic DML and cross-fitting, but their target and design differ from the controlled relevance experiment considered here. Chapter 3 states the exact structural model, nuisance construction, estimators, and inference procedure used in the simulations.

3 Methodology

3.1 Structural IVQR Model and Identification

Following the instrumental-variable quantile regression (IVQR) framework of Chernozhukov and Hansen (2005) and Chernozhukov and Hansen (2008), let Y_i denote a scalar outcome, D_i an endogenous treatment or vector of endogenous variables, X_i a vector of observed controls, and Z_i a vector of instruments. The structural quantile function is denoted by $q(u, d, x)$, where $u \in (0, 1)$ is the structural rank. For a fixed treatment state d , $q(\tau, d, x)$ gives the conditional τ -quantile of the potential outcome obtained by fixing treatment at d .

Under the IVQR assumptions on instrument exogeneity, treatment selection, and rank similarity, the observed data admit the representation

$$Y_i = q(U_i, D_i, X_i), \quad U_i \mid X_i, Z_i \sim \text{Uniform}(0, 1), \quad (1)$$

where $U_i := U_{D_i i}$ is the realized structural rank. Rank similarity allows the treatment-state-specific ranks to differ while restricting their conditional distributions; rank invariance is the stronger special case in which the rank is unchanged across treatment states. The strict monotonicity of $q(u, d, x)$ in u links this representation to structural quantiles.

Endogeneity means that the realized treatment D_i may depend on the structural rank U_i . Consequently, the observed conditional quantile of Y_i given D_i and X_i generally differs from the structural quantile obtained by externally fixing treatment. IVQR is designed to recover the latter object. For two treatment states d_1 and d_0 , the structural quantile treatment effect at

covariates x is

$$q(\tau, d_1, x) - q(\tau, d_0, x).$$

For a fixed quantile index $\tau \in (0, 1)$, the DML-IVQR framework of Chen et al. (2021) uses a linear-in-parameters structural quantile specification,

$$q(\tau, D_i, X_i) = c_0(\tau) + D_i' \alpha_0(\tau) + X_i' \beta_0(\tau). \quad (2)$$

The parameter of interest is $\alpha_0(\tau)$, the structural treatment coefficient at quantile τ . The vector $\beta_0(\tau)$ contains the coefficients on the observed controls. The term $c_0(\tau)$ is simply the intercept of the structural quantile equation: it determines the baseline structural quantile when the treatment and nonconstant controls are set to zero. It is written explicitly here rather than being absorbed into the control vector. In the particular Monte Carlo DGP used in Section 3.5, the structural outcome equation has constant term 1, so the true structural intercept is $c_0(\tau) = 1$ for every τ . Keeping $c_0(\tau)$ symbolic in this chapter allows the methodology to be stated in its general form.

The key IVQR restriction follows directly from the structural-rank representation. Because $q(u, d, x)$ is strictly increasing in u ,

$$Y_i \leq q(\tau, D_i, X_i) \iff U_i \leq \tau.$$

Conditional on X_i and Z_i , the structural rank is uniform, so

$$\Pr(U_i \leq \tau \mid X_i, Z_i) = \tau.$$

Substituting the linear structural quantile specification therefore gives

$$\Pr[Y_i \leq c_0(\tau) + D_i' \alpha_0(\tau) + X_i' \beta_0(\tau) \mid X_i, Z_i] = \tau. \quad (3)$$

Throughout the remainder of the thesis, $\mathbf{1}\{A\}$ denotes the indicator function: it equals one when statement A is true and zero otherwise. The probability restriction can therefore be written as the conditional moment

$$\mathbb{E}[\tau - \mathbf{1}\{Y_i \leq c_0(\tau) + D_i' \alpha_0(\tau) + X_i' \beta_0(\tau)\} \mid X_i, Z_i] = 0. \quad (4)$$

This equation is the central population restriction used by IVQR. At the true structural parameters, the indicator has conditional expectation τ . A candidate treatment coefficient that is inconsistent with the structural quantile restriction will generally produce a nonzero moment. Instrument validity makes these moments legitimate; instrument relevance determines

how strongly they change when the candidate treatment coefficient moves away from the truth. Hence validity and identification are distinct. A valid instrument can still contain too little treatment variation to distinguish competing values of $\alpha_0(\tau)$ precisely. The next subsection shows how DML-IVQR constructs a nuisance-robust version of this candidate-specific moment.

3.2 Orthogonal Score and Nuisance Estimation

Fix a quantile τ and consider a candidate value a for the structural treatment coefficient $\alpha_0(\tau)$. IVQR evaluates such candidate values one at a time. For a given a , subtracting $D_i' a$ from the outcome leaves an adjusted outcome whose conditional τ -quantile must be explained by an intercept and the controls. The first nuisance-estimation problem therefore profiles these quantities conditional on the candidate a .

Define the augmented control vector

$$\tilde{X}_i = (1, X_i')',$$

where the first component is the constant and the remaining components are the p nonconstant controls. Quantile regression measures fit with the check-loss function

$$\rho_\tau(u) = u[\tau - \mathbf{1}\{u < 0\}]. \quad (5)$$

Here u is a generic scalar residual argument of the check-loss function. The check loss is asymmetric: observations above and below the candidate quantile receive weights determined by τ . Minimizing this loss therefore fits the conditional τ -quantile rather than the conditional mean.

Following Chen et al. (2021), for each candidate a , let $c(a)$ denote the profiled intercept and let $\beta(a)$ denote the profiled vector of control slopes. They are defined jointly by

$$\begin{pmatrix} c(a) \\ \beta(a) \end{pmatrix} = \arg \min_{c, b} E[\rho_\tau(Y_i - D_i' a - c - X_i' b)]. \quad (6)$$

Here $c \in \mathbb{R}$ is the optimization variable for the intercept and $b \in \mathbb{R}^p$ is the optimization vector for the control slopes. Their minimizers are $c(a)$ and $\beta(a)$. These are candidate-specific nuisance parameters: changing a changes the adjusted outcome $Y_i - D_i' a$, and therefore generally changes the fitted intercept and control coefficients. At the true candidate $a = \alpha_0(\tau)$, maintained uniqueness gives $c(a) = c_0(\tau)$ and $\beta(a) = \beta_0(\tau)$.

$$E[\{\tau - \mathbf{1}[Y_i \leq D_i' a + c(a) + X_i' \beta(a)]\} \tilde{X}_i] = 0. \quad (7)$$

The moment above is the first-order quantile-regression condition for the profiled intercept and controls. It says that, at the fitted candidate-specific quantile, the quantile residual is orthogonal to the augmented controls.

With the candidate treatment effect and profiled nuisance coefficients fixed, define the candidate residual

$$\epsilon_i(a) = Y_i - D_i'a - c(a) - X_i'\beta(a). \quad (8)$$

The residual equals zero at the candidate fitted quantile. The local behavior of a quantile moment depends on how much probability mass lies near this zero boundary. This is why the residual density at zero enters the orthogonalization. Let $f_{\epsilon(a)}(0 \mid X_i, Z_i)$ be its conditional density at zero.

Suppose $Z_i \in \mathbb{R}^{d_z}$ and $X_i \in \mathbb{R}^p$, so $\tilde{X}_i \in \mathbb{R}^{p+1}$. The next step removes from the instruments the component that is locally associated with the controls. Because quantile moments change according to the residual density at zero, this residualization is density weighted rather than an ordinary unweighted linear projection. Define

$$\begin{aligned} M(a) &= \mathbb{E}[Z_i \tilde{X}_i' f_{\epsilon(a)}(0 \mid X_i, Z_i)], \quad M(a) \in \mathbb{R}^{d_z \times (p+1)}, \\ J(a) &= \mathbb{E}[\tilde{X}_i \tilde{X}_i' f_{\epsilon(a)}(0 \mid X_i, Z_i)], \quad J(a) \in \mathbb{R}^{(p+1) \times (p+1)}. \end{aligned} \quad (9)$$

Assuming $J(a)$ is nonsingular, set

$$\delta(a) = M(a)J(a)^{-1}, \quad \delta(a) \in \mathbb{R}^{d_z \times (p+1)}, \quad (10)$$

Thus $J(a)$ describes density-weighted variation in the augmented controls, while $M(a)$ describes their density-weighted relationship with the instruments. The matrix $\delta(a) = M(a)J(a)^{-1}$ contains the coefficients of the corresponding population residualization. Partition this matrix as $\delta(a) = [d_0(a), \Gamma(a)]$, where $d_0(a) \in \mathbb{R}^{d_z}$ contains the residualization intercepts and $\Gamma(a) \in \mathbb{R}^{d_z \times p}$ contains the slopes on the nonconstant controls. The corresponding population residualized instrument is

$$\Psi_i(a) = Z_i - \delta(a)\tilde{X}_i = Z_i - d_0(a) - \Gamma(a)X_i. \quad (11)$$

The vector $\Psi_i(a)$ is therefore the instrument after removing the density-weighted component associated with the intercept and controls. It is this residualized instrument, rather than the raw Z_i , that enters the theoretical orthogonal score. By construction, $\mathbb{E}[(Z_i - \delta(a)\tilde{X}_i)\tilde{X}_i' f_{\epsilon(a)}(0 \mid X_i, Z_i)] = 0$; this is not an ordinary unweighted residual or a conventional first-stage coefficient.

The candidate-specific nuisance vector collects the quantities required to construct the score.

$$\eta(a) = \begin{pmatrix} c(a) \\ \beta(a) \\ \text{vec}\{\delta(a)\} \end{pmatrix}, \quad \eta_0 = \eta(\alpha_0(\tau)) = \begin{pmatrix} c_0(\tau) \\ \beta_0(\tau) \\ \text{vec}\{\delta(\alpha_0(\tau))\} \end{pmatrix}. \quad (12)$$

Vectorization only stacks the residualization matrix. The target remains $\alpha_0(\tau)$; $c(a)$ and $\beta(a)$ profile the outcome nuisance, while $\delta(a)$ residualizes the instruments.

Following Chen et al. (2021), the orthogonal score combines two objects: the candidate quantile residual and the density-weighted residualized instrument.

$$\begin{aligned} g_\tau(W_i; a, \eta) &= [\tau - \mathbf{1}\{Y_i \leq D_i' a + c + X_i' \beta\}] (Z_i - \delta \tilde{X}_i), \\ g_\tau(W_i; a, \eta(a)) &= [\tau - \mathbf{1}\{Y_i \leq D_i' a + c(a) + X_i' \beta(a)\}] \Psi_i(a), \end{aligned} \quad (13)$$

where $W_i = (Y_i, D_i, X_i, Z_i)$. The first factor asks whether the observed outcome lies below the candidate structural quantile. The second factor supplies instrument variation after removing the control-related component. At the true treatment coefficient and true nuisance values, their population product has expectation zero.

At $a = \alpha_0(\tau)$ and $\eta = \eta_0$, the score satisfies

$$\mathbb{E}[g_\tau(W_i; \alpha_0(\tau), \eta_0)] = 0. \quad (14)$$

Ordinary plug-in estimation of high-dimensional nuisances can contaminate a target moment. DML therefore uses a score with the stronger local property that its population expectation is insensitive, to first order, to small perturbations of the nuisance vector at the truth. Formally,

$$\partial_\eta \mathbb{E}[g_\tau(W_i; \alpha_0(\tau), \eta)]|_{\eta=\eta_0} = 0. \quad (15)$$

Here ∂_η denotes differentiation with respect to the nuisance vector. Orthogonality does not mean that nuisance estimation is error free. It means that sufficiently small nuisance-estimation errors do not enter the target moment at first order.

The cancellation can be seen directly. First consider perturbations of the profiled intercept and control slopes. To show the cancellation for the intercept and slopes, write $\epsilon_{0i} = \epsilon_i(\alpha_0(\tau))$ and define

$$\begin{aligned} M_0 &= \mathbb{E}[Z_i \tilde{X}_i' f_{\epsilon_0}(0 | X_i, Z_i)], \\ J_0 &= \mathbb{E}[\tilde{X}_i \tilde{X}_i' f_{\epsilon_0}(0 | X_i, Z_i)], \\ \delta_0 &= M_0 J_0^{-1}. \end{aligned} \quad (16)$$

At the truth, differentiation with respect to the fitted intercept and control slopes then gives

$$\begin{aligned}\partial_{(c,\beta')} E[g_\tau(W_i; \alpha_0(\tau), \eta)]|_{\eta=\eta_0} &= -E[(Z_i - \delta_0 \tilde{X}_i) \tilde{X}_i' f_{\epsilon_0}(0 | X_i, Z_i)] \\ &= -[M_0 - \delta_0 J_0] = 0.\end{aligned}\tag{17}$$

The density appears because observations near the boundary can switch sides. The choice $\delta_0 = M_0 J_0^{-1}$ makes the residualized instrument density-weighted orthogonal to $\tilde{X}_i$, cancelling first-order changes in the intercept and slopes.

A similar argument applies to the instrument-residualization nuisance. For instrument component $j = 1, \dots, d_z$, let δ_j denote the corresponding row of δ . Then

$$\partial_{\delta_j} E[g_{\tau,j}(W_i; \alpha_0(\tau), \eta)]|_{\eta=\eta_0} = -E[\{\tau - \mathbf{1}[Y_i \leq D_i' \alpha_0(\tau) + c_0(\tau) + X_i' \beta_0(\tau)]\} \tilde{X}_i'] = 0.\tag{18}$$

An error in δ is multiplied by the quantile residual, whose moment with $\tilde{X}_i$ is zero by (7), so its first-order effect also vanishes. This construction permits sufficiently accurate regularized nuisances to support inference on the low-dimensional target (Chernozhukov et al. 2018; Chen et al. 2021).

Orthogonality protects the score against small nuisance-parameter errors, but it does not create identification or add information about the treatment effect. In particular, it does not make a weak instrument strong and does not guarantee that the data contain enough information about $\alpha_0(\tau)$. If the instruments contain little information about the endogenous treatment, the moment may also contain little information for distinguishing nearby values of a . IVQR moment validity comes from the structural assumptions and instrument exogeneity; nuisance robustness comes from orthogonalization. These are different requirements.

3.3 Estimators and Executable Implementation

Throughout this methodology section, N denotes the sample size. Section 3.5 uses n for the corresponding Monte Carlo design sample size. Section 3.2 defined population nuisances. When controls are numerous relative to N , DML-IVQR estimates their sample counterparts with ℓ_1 regularization under approximate sparsity: a relatively small group has substantial predictive content while other coefficients are zero or small.

Fix τ and a candidate value a . Nuisance estimation is repeated whenever a relevant candidate is evaluated. In the executable implementation used here, the response for the first nuisance step is $Y_i - D_i' a$. The formula adds an intercept, so let b_0 denote that intercept and let $b = (b_1, \dots, b_p)'$ contain only the slopes on X_i . Thus, $(\hat{b}_0(a), \hat{\beta}(a))$ estimates $(c(a), \beta(a))$.

The implementation-specific notation for the later instrument regression is defined below and is kept distinct from the full population matrix $\delta(a) = [d_0(a), \Gamma(a)]$. The candidate-specific quantile-Lasso estimates are

$$\begin{pmatrix} \widehat{b}_0(a) \\ \widehat{\beta}(a) \end{pmatrix} = \arg \min_{b_0, b} \left\{ \sum_{i=1}^N \rho_\tau(Y_i - D_i' a - b_0 - X_i' b) + \lambda_0(a) |b_0| + \sum_{k=1}^p \lambda_k(a) |b_k| \right\}. \quad (19)$$

The first term measures fit at quantile τ . The second places a separate penalty on each coefficient, including the intercept; there is no post-Lasso refit. The loss is an unscaled sum because this is the scaling used by the fitted quantile-regression routine. The implementation uses the pivotal quantile-Lasso calibration developed by Belloni and Chernozhukov (2011), following the executable route preserved by Chen et al. (2021). Let $s_k = (N^{-1} \sum_i X_{ik}^2)^{1/2}$ and, for $r = 1, \dots, 1000$, draw independent $U_{ir} \sim \text{Uniform}(0, 1)$. The implemented penalty is

$$\begin{aligned} T_r &= \max_{1 \leq k \leq p} \left| \frac{\sum_{i=1}^N X_{ik} [\tau - \mathbf{1}\{U_{ir} < \tau\}]}{s_k \sqrt{\tau(1-\tau)}} \right|, \\ \lambda(a) &= 2 \widehat{Q}_{0.9}(\{T_r\}_{r=1}^{1000}) \sqrt{\tau(1-\tau)} (1, s_1, \dots, s_p)'. \end{aligned} \quad (20)$$

Here $k = 1, \dots, p$ indexes the control variables, $r = 1, \dots, 1000$ indexes the simulated pivotal draws, T_r is the simulated pivotal statistic in draw r , and $\widehat{Q}_{0.9}(\{T_r\}_{r=1}^{1000})$ denotes their empirical 90th percentile. The $p + 1$ entries of $\lambda(a)$ correspond respectively to the intercept and the p control slopes. Artificial Uniform variables are useful because the indicator at the correct quantile has a known Bernoulli structure. The simulated statistics approximate how large random correlations between the controls and a quantile score can be without a true signal. The simulated 0.9 quantile sets the reference level for these random correlations, while the factor 2 scales the resulting penalty according to the implemented pivotal rule. A fresh set of Uniform draws is generated every time the penalty routine is called. Hence, candidate evaluations on the same observed sample can use different pivotal draws. This is penalty-construction randomness, not sample splitting. The frozen main simulation uses this rule rather than its older five-fold cross-validation alternative.

The fitted quantile regression produces the candidate residual and density weight

$$\begin{aligned} \widehat{\epsilon}_i(a) &= Y_i - D_i' a - \widehat{b}_0(a) - X_i' \widehat{\beta}(a), \\ \widehat{f}_i(a) &= \phi(\widehat{\epsilon}_i(a); \bar{\epsilon}(a), s_{\bar{\epsilon}(a)}^2), \end{aligned} \quad (21)$$

Here $\bar{\epsilon}(a)$ denotes the sample mean of the N candidate residuals, $s_{\bar{\epsilon}(a)}^2$ denotes their sample variance, and $\phi(e; \mu, \sigma)$ is the normal density with mean μ and standard deviation σ . Thus, the executable simulation forms the weights by evaluating a fitted normal density at the candidate

residuals.¹ This fitted-normal quantity is an implementation-specific proxy for the theoretical conditional density $f_{\epsilon(a)}(0 \mid X_i, Z_i)$ from Section 3.2; it is not a direct conditional-density estimate at zero. Zero remains the fitted quantile boundary, and the theoretical reason for density weighting is that observations near this boundary determine local changes in the quantile moment. These proxy weights are then used in the instrument-residualization step, where they play the role assigned to density weights in the theoretical construction.

For each instrument component $j = 1, \dots, d_z$, the implementation next fits a separate square-root-density-weighted Lasso. Let $\widehat{r}_{j0}(a)$ denote its retained ordinary intercept and let $\widehat{F}_{\text{imp},j}(a)$ contain only the slopes on the transformed controls. Its transformed regression is

$$\begin{pmatrix} \widehat{r}_{j0}(a) \\ \widehat{F}_{\text{imp},j}(a) \end{pmatrix} = \arg \min_{(r_{j0}, \gamma_j)} \left\{ \sum_{i=1}^N \left[\sqrt{\widehat{f}_i(a)} Z_{ji} - r_{j0} - \sqrt{\widehat{f}_i(a)} X_i' \gamma_j \right]^2 + \mathcal{P}_{j,a}(\gamma_j) \right\}. \quad (22)$$

Here $\mathcal{P}_{j,a}(\gamma_j) = \sum_{k=1}^p \lambda_{jk}^r(a) |\gamma_{jk}|$ is the coefficient-specific ℓ_1 penalty produced by the instrument-regression routine's default iterative heteroskedastic loadings. Here $\gamma_j \in \mathbb{R}^p$ is the slope vector for instrument component j , and $k = 1, \dots, p$ indexes its control coefficients. The intercept is included but not penalized, the same controls are used as in the first nuisance step, and no post-Lasso refit is performed. After fitting the transformed regression, the code retains both the intercept and the slopes. Define $\widehat{r}_0(a) = (\widehat{r}_{10}(a), \dots, \widehat{r}_{d_z0}(a))'$ and stack the slope rows in $\widehat{F}_{\text{imp}}(a)$. The residualized instrument used in the score is

$$\begin{aligned} \widehat{\Psi}_{ji}(a) &= Z_{ji} - \widehat{r}_{j0}(a) - X_i' \widehat{F}_{\text{imp},j}(a), \\ \widehat{\Psi}_i(a) &= Z_i - \widehat{r}_0(a) - \widehat{F}_{\text{imp}}(a) X_i. \end{aligned} \quad (23)$$

Thus, the score residualizes the original, untransformed instrument. The calculation subtracts the retained intercept and the fitted contribution of the original controls. It does not divide the retained intercept by $\sqrt{\widehat{f}_i(a)}$ or use the transformed regression residual directly.

The intercept retained by the frozen transformed regression is implementation specific. It enters as an ordinary constant rather than $\sqrt{\widehat{f}_i(a)}$ times a constant regressor, so $\widehat{r}_0(a)$ is not the literal sample counterpart of $d_0(a)$ in the augmented population projection of Section 3.2. The exact derivative cancellation in Section 3.2 applies to the theoretical augmented score. The executable score follows its density-weighted slope residualization but treats the intercept differently. The results therefore characterize the preserved implementation, not a literal sample implementation of every component of the theoretical score.

1. The literal preserved call is `dnorm(e, mean(e), var(e))`. Its third argument is interpreted as a standard deviation, although the code supplies the residual sample variance.

The penalties can shrink weak coefficients to zero and stabilize both high-dimensional steps, but also create shrinkage error. The theoretical score in Section 3.2 limits its first-order effect (Chernozhukov et al. 2018). Nuisances and scores use the same sample; this preserved implementation has no sample splitting or cross-fitting.

Oracle-GMM, Full-GMM, and DML-IVQR target the same $\alpha_0(\tau)$ using the same outcome, treatment, instruments, and quantile. Each multiplies a candidate quantile score by residualized instruments, producing a moment with dimension d_z ; nuisance estimation differs.

Oracle-GMM is an infeasible benchmark because it is given the identity of the ten controls designated as relevant in the simulation DGP. Let $S_0 = \{1, \dots, 10\}$ denote this oracle control set and let $X_{S_0,i}$ collect these variables. At each candidate, the executable estimator uses an unpenalized quantile regression with an intercept:

$$\begin{pmatrix} \widehat{b}_{0,\text{orc}}(a) \\ \widehat{\beta}_{\text{orc}}(a) \end{pmatrix} = \arg \min_{b_0, b} \sum_{i=1}^N \rho_\tau(Y_i - D_i' a - b_0 - X_{S_0,i}' b). \quad (24)$$

Thus, Oracle-GMM first removes $D_i' a$ and then fits the remaining outcome at quantile τ using only this known oracle control set. No coefficient is penalized and there is no later variable-selection or refitting step. The oracle is not a practically available estimator and is not free from sampling error. It provides a benchmark for the cost of having to learn which controls matter.

Full-GMM does not receive this knowledge. It uses all available controls in the same unpenalized, candidate-specific quantile-regression problem:

$$\begin{pmatrix} \widehat{b}_{0,\text{full}}(a) \\ \widehat{\beta}_{\text{full}}(a) \end{pmatrix} = \arg \min_{b_0, b} \sum_{i=1}^N \rho_\tau(Y_i - D_i' a - b_0 - X_i' b). \quad (25)$$

Thus all 100 controls enter freely. Full-GMM does not require the unknown oracle set, but estimating many weak or irrelevant coefficients can be noisy.

Oracle-GMM and Full-GMM use the same unregularized instrument residualization with their respective control matrices. For $e \in \{\text{orc}, \text{full}\}$, let $X_{\text{orc},i} = X_{S_0,i}$ and $X_{\text{full},i} = X_i$, and let $\widehat{f}_{e,i}(a)$ be the fitted-normal proxy defined above. The code computes

$$\begin{aligned} \widehat{M}_e(a) &= \frac{1}{N} \sum_{i=1}^N Z_i X_{e,i}' \widehat{f}_{e,i}(a), & \widehat{J}_e(a) &= \frac{1}{N} \sum_{i=1}^N X_{e,i} X_{e,i}' \widehat{f}_{e,i}(a), \\ \widehat{\delta}_e(a) &= \widehat{M}_e(a) \widehat{J}_e(a)^{-1}, & \widehat{\Psi}_{e,i}(a) &= Z_i - \widehat{\delta}_e(a) X_{e,i}. \end{aligned} \quad (26)$$

This direct, unpenalized inversion includes no residualization intercept, although the quantile regression has one, and uses the original Z_i in the final residual. It therefore differs from

the augmented population orthogonalization in Section 3.2. There is no ridge adjustment or generalized inverse: a singular $\widehat{J}_e(a)$ produces a recorded numerical failure. Each benchmark multiplies $\widehat{\mathcal{P}}_{e,i}(a)$ by its candidate quantile score.

DML-IVQR uses all controls and regularizes both nuisance steps. It estimates the control coefficients by quantile Lasso, forms the density proxy, and uses regularized weighted instrument regressions. DML-IVQR-BC applies (20). Oracle and Full also use residualized IVQR moments, but estimate nuisances without ℓ_1 regularization. Unlike their matrix calculation, DML retains a residualization intercept. The comparison covers a known oracle set without regularization, all controls without regularization, and all controls with regularized nuisance estimation. It does not isolate every implementation difference but keeps the structural model, target, and instruments common.

3.4 GMM Criterion and Weak-Identification-Robust Inference

Let $e \in \{\text{orc}, \text{full}, \text{dml}\}$ index the estimator and $\widehat{g}_{e,i}(a) \in \mathbb{R}^{d_z}$ its observation-level score. Its sample average is

$$\widehat{g}_{e,N}(a) = \frac{1}{N} \sum_{i=1}^N \widehat{g}_{e,i}(a). \quad (27)$$

This vector measures the sample restriction at a . The code measures its variation with the uncentered second moment

$$\widehat{\Sigma}_e(a) = \frac{1}{N} \sum_{i=1}^N \widehat{g}_{e,i}(a) \widehat{g}_{e,i}(a)'. \quad (28)$$

This matrix standardizes differently scaled and correlated components. It uses denominator N , no finite-sample correction, and no sample-mean centering. At the truth, its probability limit equals the score covariance because the population moment is zero.

The common candidate-specific GMM criterion is

$$W_{e,N}(a) = N \widehat{g}_{e,N}(a)' \widehat{\Sigma}_e(a)^{-1} \widehat{g}_{e,N}(a). \quad (29)$$

This equals the code's calculation from the unnormalized score sum and sum of outer products. Smaller values indicate moments closer to zero after standardization. The direct inverse has no ridge or generalized inverse; failure or a nonfinite result is recorded. Suppressing e gives $W_N(a)$. Let $\mathcal{A}$ denote the set of candidate treatment-effect values over which the GMM

profile is evaluated; its finite numerical specification in the Monte Carlo experiment is given in Section 3.7. The point estimator is

$$\widehat{\alpha}_e(\tau) = \arg \min_{a \in \mathcal{A}} W_{e,N}(a). \quad (30)$$

The minimum need not be zero. The implementation minimizes over the finite profile in Section 3.7, selects the first exact tie, and gives no point estimate to an incomplete profile.

Point estimation retains the minimizer; inference tests each a separately. Under the regularity conditions for the corresponding theoretical candidate-specific IVQR score, the null statistic has the asymptotic reference distribution

$$W_{e,N}(a) \xrightarrow{d} \chi_{d_z}^2, \quad d_z = \dim(Z_i). \quad (31)$$

The degrees of freedom equal the number of moment components. The preserved Monte Carlo implementation applies this chi-square law as the asymptotic reference distribution for the candidate-specific statistics of all three estimators. For DML-IVQR-BC, however, the implementation-specific residualization intercept described in Section 3.3 is not the literal sample counterpart of the augmented theoretical intercept. This thesis therefore does not establish a separate limit theorem for that implementation detail. Following Chernozhukov and Hansen (2008), the test evaluates the hypothesized value directly rather than using a normal approximation for $\widehat{\alpha}_e(\tau)$. At level γ , reject when $W_{e,N}(a) > c_{1-\gamma, d_z}$. Here $\gamma \in (0, 1)$ is the test significance level, and $c_{1-\gamma, d_z}$ is the $(1 - \gamma)$ -quantile of a chi-square distribution with d_z degrees of freedom. This test-level γ is distinct from the instrument-regression slope vector γ_j used in Section 3.3.

Inverting these candidate-specific tests gives

$$C_{e,1-\gamma}(\tau) = \{a \in \mathcal{A} : W_{e,N}(a) \leq c_{1-\gamma, d_z}\}. \quad (32)$$

Test inversion retains every candidate not rejected by its own moment test. Because the profile can cross the cutoff repeatedly, the region may be connected, disconnected, wide, or empty. A finite numerical domain describes only acceptance within that domain and cannot establish global unboundedness.

The underlying Chernozhukov–Hansen test-inversion approach can retain asymptotic validity under weak, partial, or failed identification, subject to its regularity conditions, but need not be informative. Weak instruments can leave many candidates below the cutoff. Because

the implementation uses an asymptotic chi-square critical value, finite-sample calibration remains empirical. This differs from the exact conditional simulation method of Chernozhukov et al. (2009).

3.5 Monte Carlo Design and Instrument-Relevance Extension

Sections 3.1–3.4 use N as generic sample-size notation, while Sections 3.5–3.7 and Chapter 4 use n for the Monte Carlo design sample size.

The Monte Carlo experiment builds on the high-dimensional simulation design of Chen et al. (2021), with $p = 100$ observed controls. For each observation i , the structural disturbances satisfy

$$\begin{pmatrix} u_i \\ \epsilon_i \end{pmatrix} \sim \mathcal{N}\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 & 0.3 \\ 0.3 & 1 \end{pmatrix}\right). \quad (33)$$

The disturbance ϵ_i enters the treatment equation, whereas u_i enters the outcome through $D_i u_i$. Their positive correlation therefore generates endogeneity.

The remaining primitives are mutually independent and independent of the disturbance pair in Equation (33), and they satisfy

$$x_{ji} \stackrel{\text{ind}}{\sim} \mathcal{N}(0, 1), \quad j = 1, \dots, 100, \quad z_{1i}, z_{2i}, v_{1i}, v_{2i} \stackrel{\text{ind}}{\sim} \mathcal{N}(0, 1). \quad (34)$$

Observations are independent across i . The observed controls are

$$X_{ji} = \Phi(x_{ji}), \quad j = 1, \dots, 100, \quad (35)$$

The observed control vector is $X_i = (X_{1i}, \dots, X_{100i})'$. Here Φ is the standard-normal cumulative distribution function, so $X_{ji} \sim \mathcal{U}(0, 1)$ independently across controls.

The executable design constructs two instruments as²

$$Z_{1i} = z_{1i} + v_{1i} + X_{2i} + X_{3i} + X_{4i}, \quad (36)$$

$$Z_{2i} = z_{2i} + v_{2i} + X_{7i} + X_{8i} + X_{9i} + X_{10i}. \quad (37)$$

2. The displayed Monte Carlo equations in Chen et al. (2021) write the corresponding control terms as x_j . The released simulation code instead uses the transformed controls $X_j = \Phi(x_j)$. The simulations in this thesis follow that executable implementation.

Thus $Z_i = (Z_{1i}, Z_{2i})'$, so the number of instrument components is $d_z = 2$. The z components enter the treatment index directly, the included controls induce dependence between instruments and controls, and the v components add independent noise. The executable baseline treatment is

$$D_i = \Phi(z_{1i} + z_{2i} + \epsilon_i). \quad (38)$$

The treatment is continuous and lies in $(0, 1)$.

The structural outcome is generated according to

$$Y_i = 1 + D_i + 5 \sum_{j=1}^7 X_{ji} + D_i u_i. \quad (39)$$

The outcome contains $X_{1i}, \dots, X_{7i}$; together with the controls in the instruments, $X_{1i}, \dots, X_{10i}$ form the ten-control oracle set. For $d \in (0, 1)$, Equation (39) implies

$$Y_i(d) = 1 + 5 \sum_{j=1}^7 X_{ji} + d(1 + u_i). \quad (40)$$

Because u_i is independent of X_i and standard normal,

$$Q_{Y_i(d)|X_i}(\tau) = 1 + 5 \sum_{j=1}^7 X_{ji} + d [1 + \Phi^{-1}(\tau)]. \quad (41)$$

Here Φ^{-1} denotes the standard-normal quantile function, i.e. the inverse of the standard-normal cumulative distribution function Φ . The structural quantile treatment effect is therefore

$$\alpha_0(\tau) = 1 + \Phi^{-1}(\tau). \quad (42)$$

In particular, $\alpha_0(0.50) = 1$.

The original high-dimensional design of Chen, Huang, and Tien fixes instrument relevance. The thesis instead requires a controlled experiment that weakens the excluded-instrument signal while preserving other economically important features. Simply replacing $z_{1i} + z_{2i}$ by $\kappa(z_{1i} + z_{2i})$ would also reduce latent-index variance. Since $D_i = \Phi(d_i^*)$, this would change the marginal treatment distribution and, through the outcome equation, confound the interpretation of performance differences. An independent Gaussian component compensates for this lost variance.

The extension varies excluded-instrument relevance through $\kappa \in [0, 1]$. Let $w_i \sim \mathcal{N}(0, 1)$ be independent of all preceding primitives. The latent index and treatment are

$$d_i^*(\kappa) = \kappa(z_{1i} + z_{2i}) + \sqrt{2(1 - \kappa^2)} w_i + \epsilon_i, \quad (43)$$

$$D_i(\kappa) = \Phi(d_i^*(\kappa)). \quad (44)$$

This symmetric loading is the sole structural modification to the executable baseline. The reason for the compensating coefficient is that independent standard-normal z_{1i} and z_{2i} give their sum variance two, whereas $\kappa(z_{1i} + z_{2i})$ has variance $2\kappa^2$. An independent standard-normal w_i with coefficient $c(\kappa)$ restores the original variance when $2\kappa^2 + c(\kappa)^2 = 2$. Taking the nonnegative root gives $c(\kappa) = \sqrt{2(1 - \kappa^2)}$. The coefficient therefore follows from variance preservation rather than an arbitrary normalization.

At $\kappa = 1$, the coefficient on w_i is zero and

$$D_i(1) = \Phi(z_{1i} + z_{2i} + \epsilon_i), \quad (45)$$

which reproduces Equation (38) and provides a baseline equivalence check. As κ decreases, the direct contribution of the excluded z components falls and independent w_i supplies more of the same index variance. Conceptually, at $\kappa = 0$ the z components disappear completely while index variance and the endogeneity channel remain unchanged. The experiment stops at $\kappa = 0.10$ to study progressively weak but nonzero designed relevance; this setting does not itself establish formal non-identification. Independence and unit variances imply

$$\text{Var}[d_i^*(\kappa)] = 2\kappa^2 + 2(1 - \kappa^2) + 1 = 3. \quad (46)$$

Thus $d_i^*(\kappa) \sim \mathcal{N}(0, 3)$ for every κ , preserving the marginal distribution of $D_i(\kappa)$. This distribution is not uniform because Φ is applied to an index with variance three.

Instrument relevance changes because

$$\text{Cov}(Z_{1i}, d_i^*(\kappa)) = \text{Cov}(Z_{2i}, d_i^*(\kappa)) = \kappa. \quad (47)$$

Because all other instrument components are independent of $d_i^*(\kappa)$, each instrument–index covariance falls as κ decreases. Only the excluded z component of each observed instrument enters the index directly, so this covariance changes one-for-one with κ . The parameter governs relevance but is not a complete measure of quantile-specific IVQR identification. Because the observed instruments also contain controls and independent noise, κ is neither a conventional first-stage coefficient nor a universal weak-IV statistic. First-stage F and partial R^2 are reported later as descriptive calibration diagnostics; no Stock–Yogo or $F < 10$ classification is imposed on IVQR. Endogeneity remains fixed because u_i is independent of z_{1i} , z_{2i} , and w_i while its covariance with ϵ_i is unchanged:

$$\text{Cov}(u_i, d_i^*(\kappa)) = \text{Cov}(u_i, \epsilon_i) = 0.3. \quad (48)$$

Joint normality and the invariant variance and covariance in Equations (46) and (48) keep the joint distribution of $(u_i, d_i^*(\kappa))$ unchanged. The controls, observed-instrument construction, structural outcome equation, induced marginal outcome distribution, and structural quantile effect are also unchanged. In particular, the 100-dimensional control distribution, construction of Z_{1i} and Z_{2i} , two instrument moments, estimator definitions, and inference procedure remain fixed. Together with $\text{Var}(d_i^*) = 3$ and $\text{Cov}(u_i, d_i^*) = 0.3$, these invariances make the exercise a relevance stress test: the intentionally changing feature is the excluded z components' contribution to treatment.

Within each replication, the same primitive draw of $u_i, \epsilon_i, X_i, z_{1i}, z_{2i}, v_{1i}, v_{2i}$, and w_i is reused across all four κ values. This common-random-number construction makes matched comparisons more informative because strength differences are not generated by unrelated samples. It does not eliminate Monte Carlo uncertainty.

3.6 Simulation Design and Estimator Comparison

The completed Monte Carlo experiment uses the design dimensions

$$\begin{aligned} n &\in \{500, 1000\}, & \tau &\in \{0.10, 0.25, 0.50, 0.75, 0.90\}, \\ \kappa &\in \{1, 0.50, 0.25, 0.10\}, & R &= 500. \end{aligned} \tag{49}$$

Here n is the Monte Carlo sample size, τ is the quantile index, κ is the instrument-relevance design parameter, and $R = 500$ is the number of Monte Carlo replications in each (n, τ, κ) design. The Cartesian product contains 40 distinct (n, τ, κ) designs, each with 500 replications.

Oracle-GMM uses the known oracle control set; Full-GMM uses all controls without regularization; and DML-IVQR uses regularized high-dimensional nuisances and an orthogonalization-based score construction, with the implementation-specific residualization intercept described in Section 3.3. The design therefore has 120 design–estimator cells. Every comparison uses the truth $\alpha_0(\tau) = 1 + \Phi^{-1}(\tau)$ from Equation (42).

All estimators use the same sample within a replication. DML-IVQR uses the BC pivotal penalty and same-sample procedure in Section 3.3. Inference is conducted at 95%, and power uses the alternatives defined below.

3.7 Performance Measures and Numerical Inference

For a fixed $(n, \tau, \kappa, \text{estimator})$ cell, let $\hat{\alpha}_r(\tau)$ denote the point estimate in replication r , and let $\alpha_0(\tau)$ be the true value from Equation (42). Replications are indexed by $r = 1, \dots, R$.

Suppressing an estimator superscript, point-estimation performance is summarized by

$$\begin{aligned} \text{Bias}(\tau) &= \frac{1}{R} \sum_{r=1}^R [\alpha_0(\tau) - \widehat{\alpha}_r(\tau)], \\ \text{MAE}(\tau) &= \frac{1}{R} \sum_{r=1}^R |\widehat{\alpha}_r(\tau) - \alpha_0(\tau)|, \\ \text{RMSE}(\tau) &= \left\{ \frac{1}{R} \sum_{r=1}^R [\widehat{\alpha}_r(\tau) - \alpha_0(\tau)]^2 \right\}^{1/2}. \end{aligned} \quad (50)$$

Measures are computed separately by estimator and design cell. Positive Bias means estimates lie below the truth on average; MAE records absolute error, and RMSE gives greater weight to large errors.

When the generic statistic W_N from Chapter 3 is evaluated in the Monte Carlo experiment, N equals the design sample size n ; the additional subscript r identifies the Monte Carlo replication. With $c_{0.95}$ denoting the critical value defined below, define

$$C_r(\tau) = \mathbf{1}\{W_{N,r}(\alpha_0(\tau)) \leq c_{0.95}\}, \quad \widehat{\text{Coverage}}(\tau) = \frac{1}{R} \sum_{r=1}^R C_r(\tau). \quad (51)$$

Coverage evaluates $W_{N,r}$ directly at $\alpha_0(\tau)$ and is therefore independent of the numerical grid. Empirical size is

$$\widehat{\text{Size}}(\tau) = \frac{1}{R} \sum_{r=1}^R \mathbf{1}\{W_{N,r}(\alpha_0(\tau)) > c_{0.95}\} = 1 - \widehat{\text{Coverage}}(\tau). \quad (52)$$

The equality follows from the implemented rejection rule.

Power is evaluated at deliberately false treatment-effect values

$$a_{\Delta}(\tau) = \alpha_0(\tau) + \Delta, \quad \Delta \in \{-1.00, -0.50, -0.25, 0.25, 0.50, 1.00\}. \quad (53)$$

For each offset, empirical power is

$$\widehat{\text{Power}}(\Delta, \tau) = \frac{1}{R} \sum_{r=1}^R \mathbf{1}\{W_{N,r}(a_{\Delta}(\tau)) > c_{0.95}\}. \quad (54)$$

The statistic is evaluated directly at each false value, independently of the confidence-region grid; $\Delta = 0$ is size, not power.

The numerical confidence-set calculation proceeds by test inversion (Chernozhukov and Hansen 2008). For fixed τ and estimator, take a candidate structural effect a , re-estimate the required nuisance quantities conditional on it, construct the estimator-specific IVQR score, and compute $W_N(a)$. Under the null that a is the true effect, the asymptotic reference distribution

has two degrees of freedom because there are two instrument moment components. Collecting every candidate that passes its own test gives the confidence region. It is not chosen around the point estimate: minimizing $W_N(a)$ and retaining candidates below a critical value are distinct operations.

Conventional Wald inference centered on an estimate typically relies more directly on regular local identification. Direct testing of each hypothesized effect remains useful under weak or partial identification. This robustness concerns test validity, not narrow regions: many competing effects may survive under weak relevance.

The accepted-set measure totals all accepted components inside the primary computational domain, including disconnected components. It is neither the span between extreme accepted values nor an unrestricted confidence-region length. Because $d_z = 2$, the 95% cutoff for the statistic in Section 3.4 is

$$c_{0.95} = \chi_{2,0.95}^2 = 5.991464547 \dots \quad (55)$$

A candidate is accepted when $W_N(a) \leq c_{0.95}$ and rejected when $W_N(a) > c_{0.95}$, with the 95% cutoff 5.991464547... above. This asymptotic cutoff is not an exact finite-sample guarantee. The primary grid is

$$\begin{aligned} A_0 &= [-1, 3], & h &= 0.1, \\ G_0 &= \{a_j = -1 + (j-1)h : j = 1, \dots, 41\}, \\ &= \{-1, -0.9, \dots, 2.9, 3\}. \end{aligned} \quad (56)$$

Here $A_0 = [-1, 3]$ is the fixed primary computational domain, $h = 0.1$ is the distance between adjacent grid points, j is the grid-point index, and a_j denotes the j -th candidate treatment-effect value. The set G_0 contains all 41 candidate values. There are 41 points because the four-unit interval from -1 to 3 , evaluated in increments of 0.1 and including both endpoints, contains $4/0.1 + 1 = 41$ grid points. The original authors' main Monte Carlo implementation uses `seq(-1, 3, length=41)`. Preserving its four-unit domain and 0.10 resolution avoids choosing a new grid after observing the extension results. All five quantiles, strength settings, and estimators use this common grid, making their measures comparable without differences in grid density. The resolution is inherited, not claimed to be asymptotically optimal. All five true effects lie strictly inside the domain: the extreme values are $\alpha_0(0.10) \approx -0.282$ and $\alpha_0(0.90) \approx 2.282$, with $\alpha_0(0.50) = 1$.

The distinction between the theoretical region and its finite-grid implementation is important. The theoretical region $C = \{a : W_N(a) \leq c_{0.95}\}$ ranges over the relevant parameter

space. Computation evaluates only finitely many candidates, producing an accepted set inside A_0 , a computational search region rather than an asserted parameter space. The reported summary is therefore the grid-based accepted-set measure within the primary computational domain $A_0 = [-1, 3]$, rather than unrestricted confidence-region length.

Total confidence-region length is an intuitive informativeness summary, but numerical measurement requires localization inside a sufficiently wide search domain. The initial design considered adaptive expansion; frozen diagnostics showed that weak-relevance accepted sets could keep reaching wider boundaries. Stopping at an arbitrary endpoint and reporting that finite span would make length partly an endpoint artifact. The main comparison therefore uses fixed A_0 , while separate $A_1 = [-3, 5]$ and $A_2 = [-5, 7]$ diagnostics examine endpoint sensitivity.

For replication r , define the binary grid-acceptance indicator, accepted grid, and all-grid-points indicator by

$$\begin{aligned} I_r(a_j) &= \mathbf{1}\{W_{N,r}(a_j) \leq c_{0.95}\}, \\ C_{r,G_0} &= \{a_j \in G_0 : I_r(a_j) = 1\}, \\ F_r &= \mathbf{1}\{I_r(a_j) = 1 \text{ for every } a_j \in G_0\}. \end{aligned} \quad (57)$$

For the primary domain A_0 , define left-boundary, right-boundary, and either-boundary acceptance in replication r by

$$B_r^L = I_r(a_1), \quad B_r^R = I_r(a_{41}), \quad B_r^E = \mathbf{1}\{B_r^L = 1 \text{ or } B_r^R = 1\}. \quad (58)$$

The corresponding either-boundary acceptance rate over R Monte Carlo replications is

$$\widehat{B}^E = \frac{1}{R} \sum_{r=1}^R B_r^E. \quad (59)$$

The same definition is used for an expanded diagnostic domain, with its own left and right endpoints. A boundary is accepted when the candidate-specific test does not reject the endpoint of the finite numerical search interval. A high boundary-acceptance rate therefore means that accepted components frequently reach the edge of the examined domain and are not localized inside it. An accepted endpoint suggests that a component may continue outside the domain; a measure near four indicates limited localization within A_0 . The A_1 and A_2 diagnostics examine whether this behavior disappears. Failure to localize even within A_2 does not establish global unboundedness or failed identification. Conversely, if boundary acceptance disappears after the domain is widened, the accepted component is numerically localized within the wider

range. Persistent boundary acceptance as successively wider finite domains are examined indicates continued lack of numerical localization over those investigated ranges.

The frozen implementation computes the grid-based accepted-set measure within A_0 as the trapezoidal integral of this binary indicator:

$$M_r(A_0) = \sum_{j=1}^{40} \frac{a_{j+1} - a_j}{2} [I_r(a_j) + I_r(a_{j+1})]. \quad (60)$$

The sum runs from $j = 1$ to 40 because the 41 grid points in G_0 define 40 adjacent grid intervals $[a_j, a_{j+1}]$. The trapezoidal sum includes separated accepted components without filling rejected gaps and does not interpolate critical-value crossings. With $h = 0.10$, an adjacent interval contributes zero if both endpoints are rejected, 0.10 if both are accepted, and 0.05 if exactly one is accepted. Thus the measure ranges from zero to four and summarizes numerical acceptance, including disconnected components, without solving for exact continuous crossings of $W_N(a)$ with the cutoff. The Monte Carlo mean of F_r is the frequency with which all tested grid points are accepted, not acceptance of the real line.

Coverage evaluates the exact $\alpha_0(\tau)$, whether or not it is a grid point; power evaluates the exact $\alpha_0(\tau) + \Delta$, independently of G_0 . Grid and domain choices affect the accepted-set measure and profile point estimate, but do not mechanically determine reported coverage or power. A post-main diagnostic examines sensitivity to the finite endpoints using

$$A_0 = [-1, 3], \quad A_1 = [-3, 5], \quad A_2 = [-5, 7]. \quad (61)$$

It uses $n = 1000$, 100 saved primitive samples, all five quantiles and estimators, $\kappa \in \{1, 0.25, 0.10\}$, and spacing 0.1. Stage 1 evaluates A_1 and its nested A_0 profile; Stage 2 reloads the same samples and adds only the tails needed for A_2 . Domain comparisons therefore change candidates, not samples. The outputs record accepted measure, grid share and count, boundary acceptance, all-grid acceptance, and incremental measure.

The boundary-acceptance diagnostic follows the finite-domain interpretation of Equations (58)–(59). The bounded diagnostic ends at A_2 , while main summaries retain A_0 and direct coverage and power remain unchanged.

4 Results

The analysis proceeds in four steps: verifying the relevance ordering induced by κ and examining point-estimation accuracy; assessing finite-sample coverage; evaluating accepted-set informativeness and power; and comparing quantiles and estimators. The central question is how quickly deterioration appears, and through which dimensions, in this high-dimensional IVQR design. Reading these measures together distinguishes inaccurate estimation, poorly calibrated inference, and tests that retain too many competing effects to be informative.

4.1 Instrument Relevance and Point Estimation

Table 1 establishes that the four values of κ produce clearly ordered excluded-instrument relevance settings, using the first-stage F -statistic and partial R^2 as descriptive diagnostics.

Both diagnostics fall strongly as κ decreases. For $n = 500$, the median F -statistic falls from 101.17 at $\kappa = 1$ to 18.71 at $\kappa = 0.50$, 4.74 at $\kappa = 0.25$, and 1.22 at $\kappa = 0.10$; the corresponding endpoints for $n = 1000$ are 212.82 and 2.04. Partial R^2 is comparatively stable across sample sizes, whereas F increases with n . The design therefore varies the instruments' explanatory contribution while sample size affects the amount of information available. These diagnostics describe relative relevance within the experiment; they are not quantile-specific weak-IV thresholds.

Table 1. Instrument-strength calibration

n	κ	Median F	F : p10–p90	Median partial R^2	Partial R^2 : p10–p90
500	1.00	101.17	85.93–126.68	0.294	0.261–0.342
500	0.50	18.71	11.99–27.95	0.071	0.047–0.103
500	0.25	4.74	1.74–8.65	0.019	0.007–0.034
500	0.10	1.22	0.24–3.65	0.005	0.001–0.015
1000	1.00	212.82	185.34–239.50	0.301	0.273–0.327
1000	0.50	38.65	30.13–53.60	0.073	0.058–0.098
1000	0.25	9.85	5.34–16.15	0.020	0.011–0.032
1000	0.10	2.04	0.38–5.42	0.004	0.001–0.011

Note: Descriptive Monte Carlo strength diagnostics from the preserved 100-replication calibration. The reported dispersion ranges are the 10th and 90th percentiles. These statistics are not formal weak-instrument classification thresholds.

The two diagnostics separate signal strength from sample information. Partial R^2 falls monotonically from approximately 0.29 and 0.30 at strong relevance to 0.005 and 0.004 at the weakest setting for the smaller and larger samples, respectively. Its relative stability across sample sizes reflects a similar underlying explanatory share, while its decline across κ indicates that the designed population signal is changing. The F -statistic rises with sample size because the same explanatory share is supported by more observations and estimated more precisely. The clean ordering at both sample sizes confirms that the intervention works as designed, without supplying formal IVQR identification thresholds.

Table 2 is the central finite-sample performance table. It reports Bias, MAE, RMSE, Coverage, and the median accepted-set measure for $n = 1000$, $\tau \in \{0.10, 0.50, 0.90\}$, all four κ values, and Oracle-GMM, Full-GMM, and DML-IVQR-BC.

As instrument relevance declines, MAE and RMSE generally increase for all three estimators and across the reported quantiles. At $\tau = 0.50$, moving from $\kappa = 1$ to $\kappa = 0.10$ raises RMSE from 0.072 to 0.945 for Oracle, from 0.100 to 1.018 for Full, and from 0.104 to 0.991 for DML. Oracle deterioration shows that the phenomenon is not generated solely by high-dimensional nuisance estimation: knowing the relevant controls does not replace instrument information. Under the frozen convention $\text{Bias} = \alpha_0 - \widehat{\alpha}$, positive Bias means that estimates are below the truth on average.

The progression is visible before the weakest setting. At $\kappa = 1$, estimates are relatively concentrated, particularly at the median. At $\kappa = 0.50$, deterioration is already apparent, although useful signal remains. By $\kappa = 0.25$, MAE and RMSE have increased markedly; at $\kappa = 0.10$, errors are large across estimators. Modest Bias alongside much larger MAE and RMSE means that positive and negative replication errors partly cancel. Weak relevance therefore manifests importantly as dispersion rather than necessarily a common directional displacement.

Oracle-GMM knows the designated relevant control set and removes the high-dimensional variable-selection problem by construction. Its deterioration provides a benchmark for separating the common relevance problem from additional high-dimensional estimation difficulties. The shared loss of accuracy cannot be attributed solely to Lasso, nuisance selection, or DML regularization: all three procedures face reduced information from the excluded instruments.

4.2 Coverage and Finite-Sample Calibration

Figure 1 and the Coverage column of Table 2 show that empirical coverage is not uniformly close to the nominal level of 0.95. Coverage is evaluated directly at the true structural quantile

Table 2. Main finite-sample performance at $n = 1000$

τ	κ	Estimator	Bias	MAE	RMSE	Coverage	Median accepted-set measure
0.10	1.00	Oracle-GMM	-0.054	0.093	0.121	0.860	0.400
		Full-GMM	-0.329	0.337	0.387	0.502	0.800
		DML-IVQR-BC	-0.005	0.117	0.150	0.968	0.600
	0.50	Oracle-GMM	-0.101	0.194	0.287	0.926	1.000
		Full-GMM	-0.407	0.465	0.630	0.804	1.900
		DML-IVQR-BC	-0.159	0.306	0.482	0.948	2.225
	0.25	Oracle-GMM	-0.384	0.566	0.934	0.930	3.450
		Full-GMM	-0.773	0.914	1.242	0.932	3.500
		DML-IVQR-BC	-0.681	0.844	1.229	0.950	3.750
	0.10	Oracle-GMM	-0.747	0.975	1.376	0.946	4.000
		Full-GMM	-1.074	1.251	1.582	0.942	3.900
		DML-IVQR-BC	-0.932	1.146	1.514	0.936	4.000
0.50	1.00	Oracle-GMM	0.007	0.047	0.072	0.872	0.200
		Full-GMM	0.007	0.072	0.100	0.896	0.400
		DML-IVQR-BC	0.070	0.079	0.104	0.776	0.300
	0.50	Oracle-GMM	0.015	0.093	0.134	0.936	0.500
		Full-GMM	0.014	0.151	0.199	0.946	0.800
		DML-IVQR-BC	0.059	0.119	0.160	0.918	0.600
	0.25	Oracle-GMM	0.045	0.281	0.431	0.946	1.900
		Full-GMM	0.029	0.372	0.500	0.936	2.200
		DML-IVQR-BC	-0.003	0.306	0.448	0.948	2.025
	0.10	Oracle-GMM	-0.044	0.735	0.945	0.942	3.900
		Full-GMM	-0.012	0.840	1.018	0.948	3.800
		DML-IVQR-BC	-0.195	0.779	0.991	0.942	3.900
0.90	1.00	Oracle-GMM	0.072	0.110	0.148	0.904	0.500
		Full-GMM	0.387	0.397	0.465	0.530	1.000
		DML-IVQR-BC	0.183	0.186	0.215	0.498	0.400
	0.50	Oracle-GMM	0.211	0.295	0.540	0.940	1.575
		Full-GMM	0.584	0.629	0.871	0.834	2.450
		DML-IVQR-BC	0.226	0.260	0.396	0.844	1.100
	0.25	Oracle-GMM	0.491	0.658	1.038	0.952	3.650
		Full-GMM	0.901	1.058	1.389	0.920	3.600
		DML-IVQR-BC	0.417	0.580	0.927	0.922	3.350
	0.10	Oracle-GMM	0.653	0.948	1.350	0.964	4.000
		Full-GMM	1.050	1.236	1.583	0.948	3.900
		DML-IVQR-BC	0.519	0.831	1.191	0.940	4.000

Note: Each cell is based on 500 Monte Carlo replications. Bias follows the frozen convention ($\text{Bias} = \alpha_0 - \widehat{\alpha}$). Coverage is evaluated directly at the true structural quantile effect. The accepted-set measure is the grid-based measure within the prespecified primary computational domain ($A_0 = [-1, 3]$); it is not an unrestricted confidence-region length.

effect, independently of the numerical grid. Weak-identification robustness of the asymptotic procedure does not imply exact finite-sample coverage at the simulated sample sizes.

Because each reported coverage rate is based on $R = 500$ Monte Carlo replications, the simulation itself introduces sampling uncertainty. For an estimated rejection or coverage probability $\widehat{p}$, the Monte Carlo standard error is

$$\text{MCSE}(\widehat{p}) = \sqrt{\frac{\widehat{p}(1 - \widehat{p})}{R}}.$$

With $R = 500$, the largest possible MCSE is approximately 0.022, attained near $\widehat{p} = 0.5$; at the nominal coverage probability 0.95, the corresponding MCSE is approximately 0.010. Consequently, very small differences between nearby coverage rates should not be given substantive weight. The large deviations emphasized here are materially larger than Monte Carlo uncertainty: coverage values around 0.50, 0.63, or 0.86, for example, remain far from the nominal 0.95 benchmark relative to their simulation standard errors.

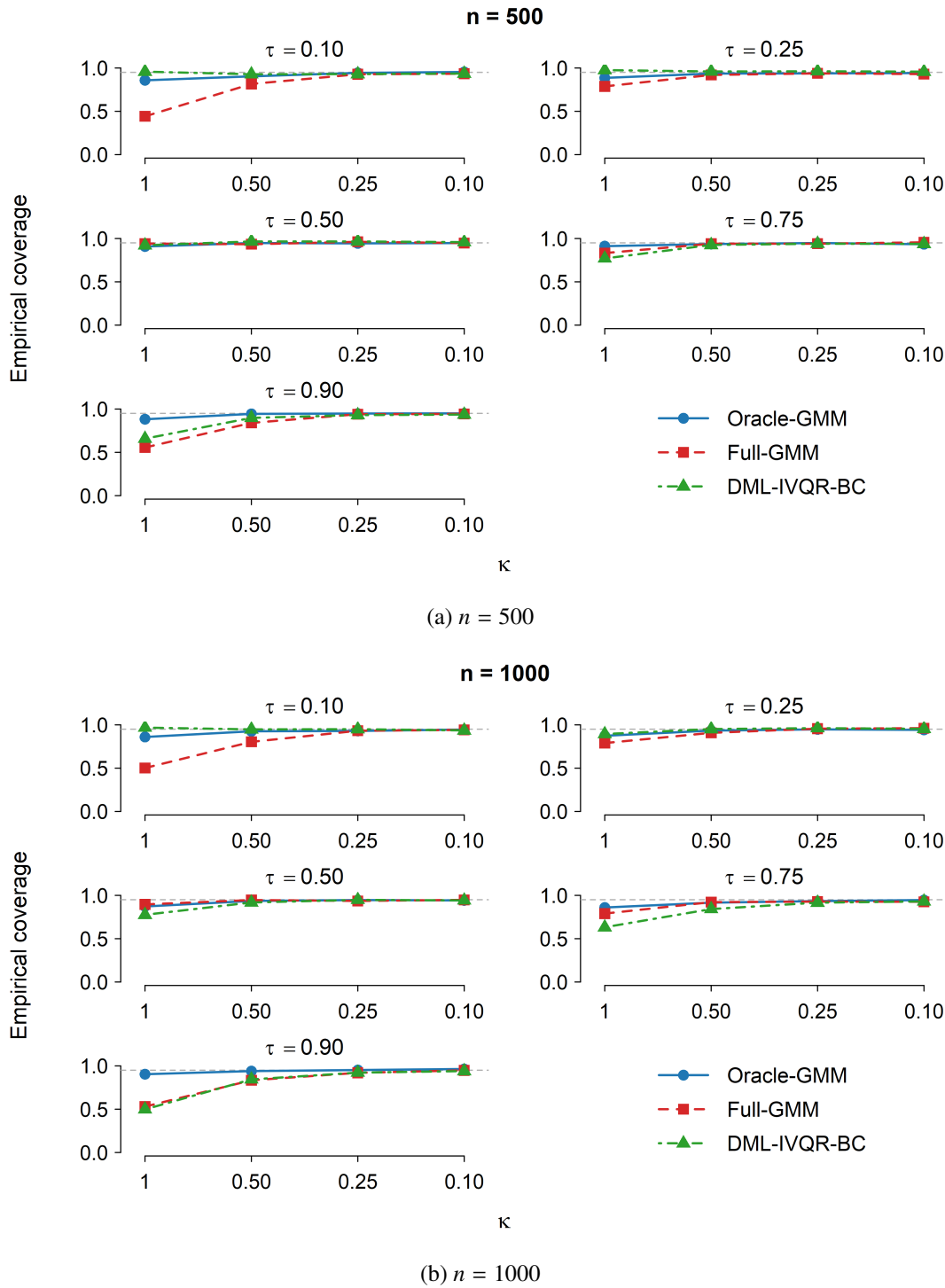
Differences of one or two percentage points warrant caution, whereas deviations of tens of percentage points from nominal coverage are far too large to be explained by the sampling uncertainty associated with $R = 500$ replications.

Distortions occur even at $\kappa = 1$ and vary substantially by estimator and quantile. For $n = 1000$, Full coverage is 0.502 at $\tau = 0.10$ and 0.530 at $\tau = 0.90$. DML coverage is 0.632 at $\tau = 0.75$ and 0.498 at $\tau = 0.90$, but 0.968 at $\tau = 0.10$. Oracle coverage is 0.860 at both $\tau = 0.10$ and $\tau = 0.75$, showing that high-dimensional nuisance estimation is not the sole explanation for undercoverage.

Coverage should be close to 0.95 when the asymptotic chi-square distribution approximates the finite-sample null distribution well. The strong-relevance differences instead reveal calibration heterogeneity across estimator–quantile combinations. Increasing the sample size does not uniformly remove these distortions: the larger sample still contains substantial undercoverage. Small differences between the two sample sizes are less informative than the persistence of large deviations.

Calibration and weak identification are distinct issues. The robust test construction avoids requiring a strongly identified point estimator in the usual way, but still uses an asymptotic chi-square cutoff. Weak-identification robustness concerns the limiting inference; calibration concerns its approximation at the simulated sample sizes. Undercoverage already present with strong relevance therefore cannot be attributed to weak identification alone.

As relevance becomes very weak, the test may reject few candidates, retaining the truth frequently while accepted sets cover most of the primary domain. Such high coverage mea-

**Figure 1.** Empirical coverage across instrument-strength designs.

sure calibration and can coexist with very low informativeness and power. An increase in coverage as relevance weakens is therefore not automatically an improvement: truth retention may rise because the procedure rejects fewer values generally. Coverage must be read alongside the evidence on exclusion of false candidates.

4.3 Confidence-Set Informativeness and Power

Figure 2 shows the median grid-based accepted-set measure within $A_0 = [-1, 3]$. Panels (a) and (b) report $n = 500$ and $n = 1000$, respectively, allowing comparison across sample sizes for the same quantiles and estimators. The plotted quantity measures acceptance inside the four-unit primary domain, not unrestricted confidence-region length.

The measure rises strongly as κ falls, and at $\kappa = 0.10$ accepted sets often occupy nearly the whole primary domain. For $n = 1000$ and $\tau = 0.50$, the Oracle, Full, and DML medians rise respectively from 0.200, 0.400, and 0.300 at $\kappa = 1$ to 3.900, 3.800, and 3.900 at $\kappa = 0.10$. Increasing n generally improves localization, but does not solve the weakest-relevance case. The accepted-set measure is deliberately defined on the fixed primary domain $A_0 = [-1, 3]$. The separate domain-expansion diagnostic is reported in Appendix A.

Small accepted-set measures under strong relevance indicate that many competing candidates are excluded. Expansion at intermediate relevance reflects a progressively less localized effect. A value near four means that almost the entire primary domain survives testing, not that the unrestricted confidence region has been shown to have length four. It is direct evidence of low localization within A_0 .

The sample-size comparison is most favorable where useful instrument signal remains: additional observations generally narrow the accepted set. At the weakest relevance, however, measures remain large even in the larger sample. More observations help, but do not fully compensate for the information deficit at these simulated sizes; two sample sizes do not establish an asymptotic conclusion.

Figure 3 reports rejection probabilities under moderate false alternatives for $n = 1000$: panel (a) uses $\Delta = -0.50$ and panel (b) uses $\Delta = +0.50$. Rejection probabilities fall strongly as κ decreases, so false candidate treatment effects survive more frequently.

The median-quantile results show a substantial loss of discrimination even at intermediate relevance. At $\tau = 0.50$ and $\Delta = -0.50$, power falls from 1.000 through 0.850 and 0.316 to 0.082 for Oracle, and from 1.000 through 0.802 and 0.312 to 0.104 for DML across the four declining relevance settings. Full power falls from 0.998 to 0.076 between the strongest and weakest settings. At the positive alternative, the corresponding weakest-relevance probabilities

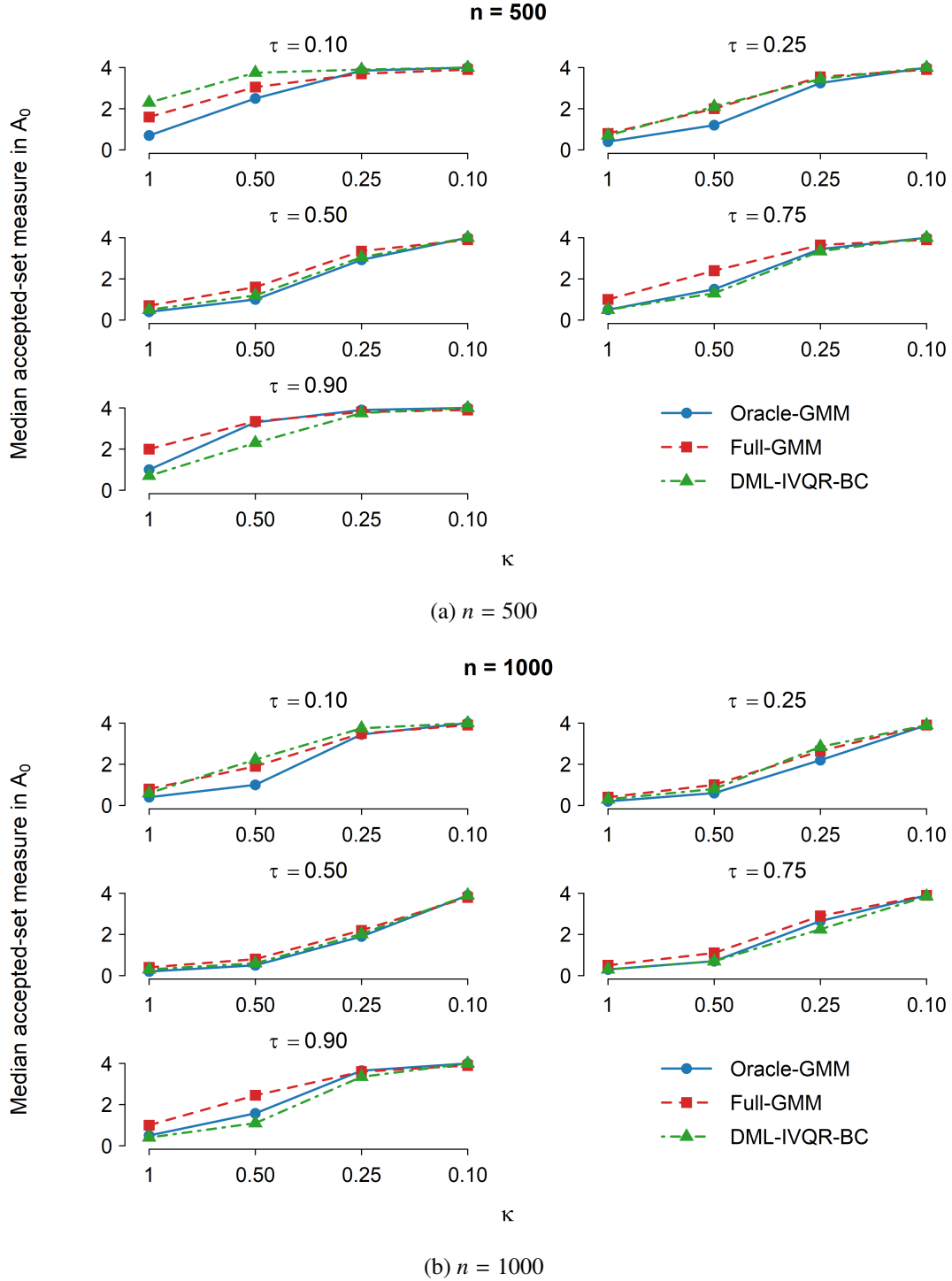


Figure 2. Grid-based accepted-set measure within the primary computational domain $A_0 = [-1, 3]$.

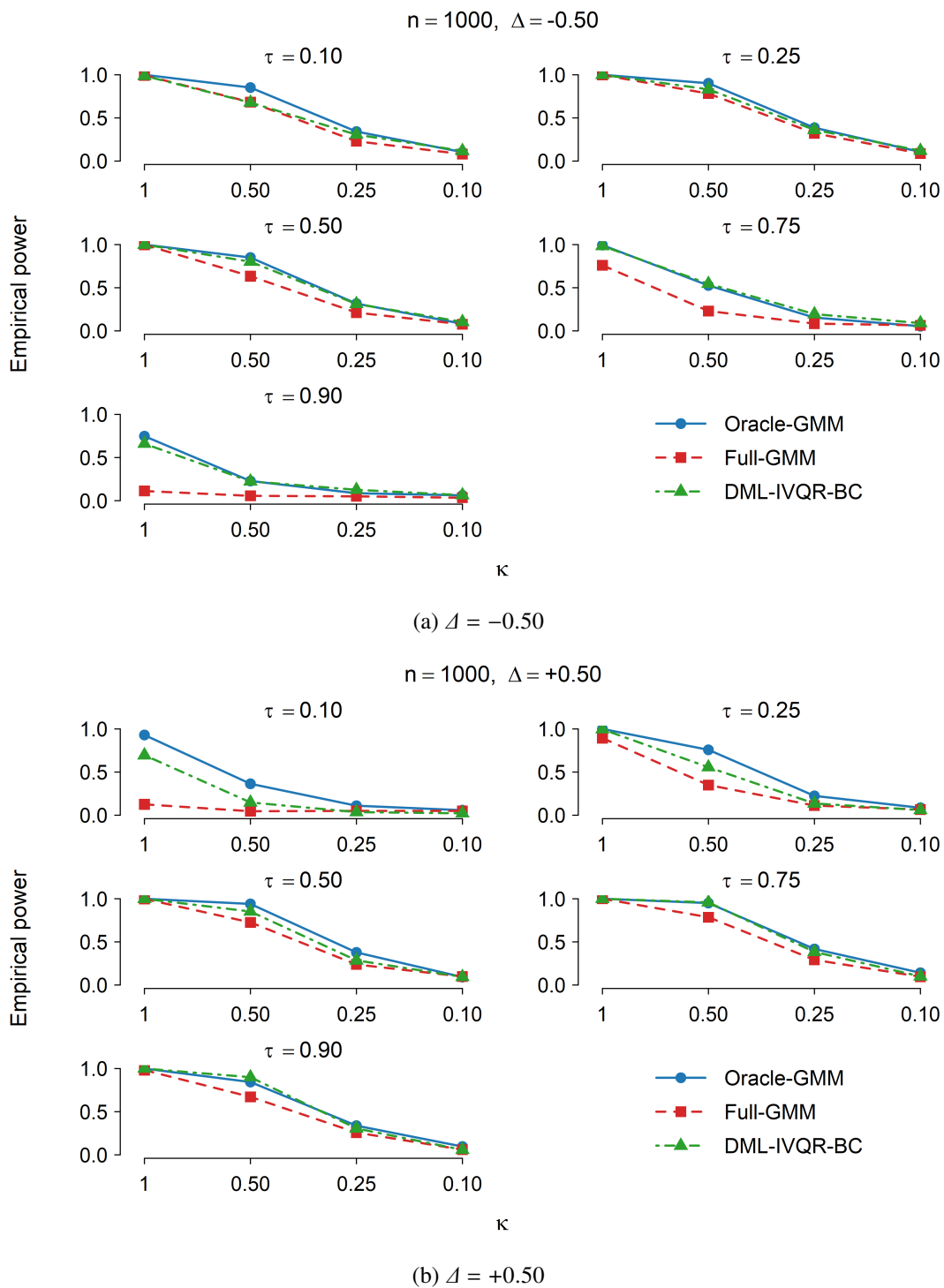


Figure 3. Rejection probability under moderate false alternatives, $n = 1000$.

are 0.092, 0.098, and 0.094. Power is the fraction of replications rejecting the specified false effect $\alpha_0(\tau) + \Delta$; its decline directly records a reduced ability to distinguish that effect from the truth. Positive and negative displacements need not have equal power because IVQR moments are nonlinear in the candidate parameter. The finite-sample criterion need not be symmetric around the truth, so equal displacements can produce different rejection probabilities. This asymmetry alone is not evidence of point-estimation bias.

Figures 2 and 3 reveal the same loss of information from complementary perspectives: accepted sets ask which candidates survive, while power asks whether specific false candidates are rejected. As relevance falls, survival expands and rejection declines. Their agreement is the clearest evidence here that weak relevance reduces inferential informativeness even with a test construction designed for weak-identification robustness.

4.4 Quantile Heterogeneity and Estimator Comparison

Taken together, Table 2 and Figures 1–3 show that weakening relevance worsens performance across quantiles, although the magnitude differs. Tail quantiles often display larger estimation errors or more severe calibration problems, but there is no universal ordering between the center and tails or between the two tails.

IVQR identification is quantile specific because the density of the structural residual near zero enters the score and information structure. Tail thresholds commonly have fewer observations effectively nearby, while nuisance estimation and residualization also vary across quantiles. The same relevance setting can therefore yield different accuracy, coverage, and power. Tail deterioration is common, but the five quantiles have no universal monotone performance ranking.

Oracle generally provides the strongest benchmark, yet deteriorates sharply as relevance weakens. Full estimates all high-dimensional controls without regularization and often has larger error or lower power. DML often improves on Full, but does not dominate it in every cell or reproduce Oracle uniformly. Its substantial coverage distortions in some quantiles also show that gains in accuracy need not imply gains in calibration. These differences cannot be attributed solely to regularization because the estimators also use different residualization conventions.

The experiment answers the research question along four dimensions. Weaker relevance substantially worsens point-estimation accuracy; accepted candidates expand and power falls, revealing reduced informativeness. These effects also occur for Oracle, so they are not solely

high-dimensional nuisance-estimation phenomena. Coverage adds a separate calibration dimension that varies across estimator and quantile and does not move mechanically with informativeness. DML can reduce some high-dimensional nuisance-estimation costs, but cannot restore information that weak instruments do not supply.

5 Conclusion

This thesis studies how systematically weakening excluded-instrument relevance affects the finite-sample performance and informativeness of weak-identification-robust inference in high-dimensional IVQR. The Monte Carlo extension varies a single relevance parameter, κ , while preserving the marginal treatment distribution, the endogeneity channel, the structural outcome equation, the true quantile treatment effects, the controls, and the observed instrument construction of the baseline design in Chen et al. (2021). Oracle-GMM, Full-GMM, and DML-IVQR-BC are evaluated on identical simulated samples across five quantiles and two sample sizes. This design isolates the consequences of progressively reducing the contribution of the excluded instruments to treatment without interpreting κ as a complete measure of quantile-specific IVQR identification.

The main result is clear. As excluded-instrument relevance declines, point estimation becomes substantially less accurate and candidate-specific inference becomes less informative. MAE and RMSE increase for all three estimators, including Oracle-GMM. At the median with $n = 1000$, for example, Oracle RMSE rises from 0.072 at $\kappa = 1$ to 0.945 at $\kappa = 0.10$, while DML-IVQR-BC rises from 0.104 to 0.991. The deterioration of the oracle benchmark shows that this pattern is not generated solely by high-dimensional nuisance estimation. Reduced instrument information directly weakens the ability of the IVQR moments to distinguish the true treatment effect from competing candidate values.

The inference results reinforce this conclusion. The grid-based accepted-set measure within the prespecified domain $A_0 = [-1, 3]$ expands sharply as relevance falls, while rejection probabilities against false treatment-effect values decline. At the weakest relevance setting, accepted values frequently occupy almost the entire primary domain and power against moderate false alternatives is low even at $n = 1000$. Larger samples improve localization and rejection power when usable instrument signal remains, but they do not eliminate the loss of informativeness at $\kappa = 0.10$. The magnitude of this deterioration also varies across quantiles: tail quantiles are often more difficult than the median, but there is no universal ordering across all estimators and relevance levels.

A separate finding concerns finite-sample calibration. Empirical coverage is not uniformly close to the nominal 0.95 level, including in some cells with the strongest relevance setting. Undercoverage also appears for Oracle-GMM, which rules out high-dimensional regularization as the sole explanation. With $R = 500$ replications, the maximum Monte Carlo standard error of a reported coverage probability is approximately 0.022, so the large coverage deviations documented in the simulations cannot be explained by Monte Carlo noise alone. These

findings indicate that the asymptotic chi-square reference distribution can be imperfectly calibrated at the simulated sample sizes. At the same time, very high coverage under weak relevance is not necessarily desirable: it can coexist with broad accepted sets and very low power against false candidates.

The estimator comparison clarifies both the usefulness and the limits of DML-IVQR. Regularized nuisance estimation often improves performance relative to unregularized Full-GMM, but DML-IVQR-BC neither dominates in every design cell nor uniformly reproduces the Oracle benchmark. Because the estimators also differ in their residualization implementation, the observed gaps cannot be interpreted as the isolated causal effect of regularization. More fundamentally, Neyman orthogonality protects the target moment against first-order nuisance-estimation error; it does not supply the identifying information lost when excluded instruments become weakly relevant.

The scope of these conclusions is deliberately limited. The analysis uses one preserved DGP, two sample sizes, one same-sample DML-IVQR implementation, a finite numerical candidate domain, and the authors' asymptotic chi-square inference procedure. It does not study alternative instrument structures, cross-fitted implementations, alternative critical-value procedures, or an empirical application. These are natural directions for further work rather than unresolved requirements of the present experiment.

The central implication is both practical and methodological. In high-dimensional IVQR, successful nuisance regularization and weak-identification-robust test construction do not by themselves guarantee precise or informative inference. When excluded instruments contain little information about treatment, estimation error can rise sharply, false structural effects may be difficult to reject, and finite-sample calibration can remain imperfect. Applied work should therefore evaluate instrument relevance, point-estimation accuracy, coverage, and inferential informativeness separately. No single measure provides sufficient evidence of reliable structural inference.

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Appendix A Numerical-Domain Diagnostics

The primary comparison domain is $A_0 = [-1, 3]$. Wider domains $A_1 = [-3, 5]$ and $A_2 = [-5, 7]$ assess sensitivity to finite search endpoints. Table A.1 retains the key frozen domain-measure comparisons; Figure A.1 summarizes the expansion pattern.

At strong relevance, the accepted set localizes more clearly and expansion adds little accepted measure. Under weak relevance, acceptance can continue toward expanded boundaries. This motivates reporting the fixed-domain accepted-set measure rather than unrestricted confidence-region length. Persistence across wider finite domains indicates poor localization over the investigated ranges, not proof of global unboundedness or failed identification.

Table A.1. Grid-domain diagnostic: accepted-set measures

n	τ	κ	Estimator	Med. A_0	Med. A_1	Med. A_2	Mean A_0	Mean A_1	Mean A_2
1000	0.10	1.00	Oracle-GMM	0.400	0.400	0.400	0.386	0.386	0.386
1000	0.10	1.00	Full-GMM	0.800	0.800	0.800	0.757	0.757	0.757
1000	0.10	1.00	DML-IVQR-BC	0.600	0.600	0.600	0.705	0.705	0.705
1000	0.10	0.25	Oracle-GMM	3.400	5.325	7.250	3.019	4.994	6.838
1000	0.10	0.25	Full-GMM	3.375	5.950	8.200	3.304	5.774	8.037
1000	0.10	0.25	DML-IVQR-BC	3.750	5.950	7.950	3.607	6.040	8.245
1000	0.10	0.10	Oracle-GMM	4.000	7.900	11.900	3.745	7.381	11.002
1000	0.10	0.10	Full-GMM	3.900	7.800	11.600	3.785	7.394	10.975
1000	0.10	0.10	DML-IVQR-BC	4.000	7.575	10.775	3.845	7.219	10.339
1000	0.25	1.00	Oracle-GMM	0.300	0.300	0.300	0.259	0.259	0.259
1000	0.25	1.00	Full-GMM	0.400	0.400	0.400	0.429	0.429	0.429
1000	0.25	1.00	DML-IVQR-BC	0.300	0.300	0.300	0.346	0.346	0.346
1000	0.25	0.25	Oracle-GMM	2.175	2.800	2.850	2.224	3.158	3.964
1000	0.25	0.25	Full-GMM	2.450	3.500	4.075	2.517	3.768	4.745
1000	0.25	0.25	DML-IVQR-BC	2.750	4.650	6.225	2.630	4.130	5.712
1000	0.25	0.10	Oracle-GMM	3.900	7.700	11.500	3.492	6.699	9.872
1000	0.25	0.10	Full-GMM	3.825	7.600	11.100	3.595	6.891	10.083
1000	0.25	0.10	DML-IVQR-BC	4.000	7.700	11.125	3.651	6.926	9.984
1000	0.50	1.00	Oracle-GMM	0.200	0.200	0.200	0.235	0.235	0.235
1000	0.50	1.00	Full-GMM	0.400	0.400	0.400	0.373	0.373	0.373
1000	0.50	1.00	DML-IVQR-BC	0.300	0.300	0.300	0.259	0.259	0.259
1000	0.50	0.25	Oracle-GMM	1.875	2.200	2.200	1.960	2.682	3.229
1000	0.50	0.25	Full-GMM	2.250	2.400	2.550	2.368	3.171	3.802
1000	0.50	0.25	DML-IVQR-BC	1.950	2.500	3.025	2.124	3.020	3.942
1000	0.50	0.10	Oracle-GMM	3.825	7.600	11.300	3.521	6.640	9.687
1000	0.50	0.10	Full-GMM	3.900	7.625	11.450	3.536	6.718	9.817
1000	0.50	0.10	DML-IVQR-BC	3.900	7.500	10.325	3.538	6.671	9.585
1000	0.75	1.00	Oracle-GMM	0.300	0.300	0.300	0.290	0.290	0.290

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TableA.1. Grid-domain diagnostic: accepted-set measures (Continued)

n	τ	κ	Estimator	Med. A_0	Med. A_1	Med. A_2	Mean A_0	Mean A_1	Mean A_2
1000	0.75	1.00	Full-GMM	0.500	0.500	0.500	0.512	0.512	0.512
1000	0.75	1.00	DML-IVQR-BC	0.300	0.300	0.300	0.279	0.279	0.279
1000	0.75	0.25	Oracle-GMM	2.825	4.200	4.850	2.628	3.945	4.990
1000	0.75	0.25	Full-GMM	2.950	4.450	5.150	2.827	4.267	5.478
1000	0.75	0.25	DML-IVQR-BC	2.325	3.300	3.900	2.402	3.595	4.788
1000	0.75	0.10	Oracle-GMM	3.975	7.750	11.700	3.670	6.999	10.357
1000	0.75	0.10	Full-GMM	3.900	7.750	11.500	3.661	7.023	10.267
1000	0.75	0.10	DML-IVQR-BC	3.900	7.350	10.375	3.554	6.691	9.736
1000	0.90	1.00	Oracle-GMM	0.500	0.500	0.500	0.519	0.519	0.519
1000	0.90	1.00	Full-GMM	1.000	1.000	1.000	1.026	1.031	1.031
1000	0.90	1.00	DML-IVQR-BC	0.400	0.400	0.400	0.454	0.454	0.454
1000	0.90	0.25	Oracle-GMM	3.650	5.850	7.850	3.381	5.625	7.675
1000	0.90	0.25	Full-GMM	3.675	6.150	8.575	3.555	6.123	8.432
1000	0.90	0.25	DML-IVQR-BC	3.450	5.400	7.300	3.114	5.249	7.374
1000	0.90	0.10	Oracle-GMM	4.000	7.900	11.800	3.712	7.241	10.771
1000	0.90	0.10	Full-GMM	3.900	7.700	11.500	3.817	7.448	11.031
1000	0.90	0.10	DML-IVQR-BC	4.000	7.700	11.600	3.644	7.042	10.395

Note: Domains are $A_0 = [-1, 3]$, $A_1 = [-3, 5]$, and $A_2 = [-5, 7]$. Diagnostics concern only these investigated numerical domains and do not establish global boundedness, unboundedness, or disconnected components.

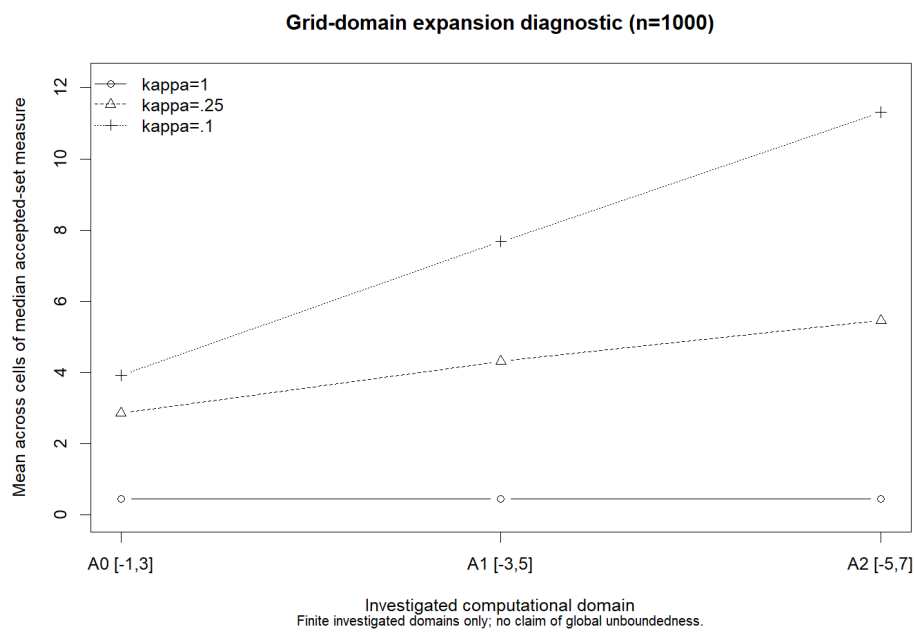


Figure A.1. Sensitivity of the accepted set to expansion of the computational domain

Note: Each plotted value is the mean across 15 estimator–quantile cells of the cell-specific median accepted-set measure for the stated κ and domain. The investigated domains are $A_0 = [-1, 3]$, $A_1 = [-3, 5]$, and $A_2 = [-5, 7]$. The diagnostic records whether the accepted set stabilizes within these numerical domains; it does not establish global boundedness or unboundedness.

Appendix B Coverage Forensic Audit

Coverage is evaluated directly at the exact true structural effect $\alpha_0(\tau)$, independently of whether that value lies on the finite candidate grid. The forensic audit checked 60,000 direct truth evaluations: 500 replications for each of 120 sample-size, quantile, instrument-strength, and estimator cells.

Wilson 95% Monte Carlo intervals exclude nominal 0.95 coverage in 54 of the 120 cells. The audit finds no missing or nonfinite true-parameter W statistics. The large coverage distortions reported in the main text are therefore neither plotting-grid artifacts nor consequences of missing-value handling. The frozen audit report and supporting diagnostics remain preserved in the repository.

Appendix C Reproducibility Note

The preserved author-replication and main-extension environment is R 3.4.3 on 64-bit Windows. The main Monte Carlo uses $R = 500$ replications per design and seed 675. The grid-domain diagnostic also uses seed 675 and reuses saved primitives in its second stage. The separate instrument-strength calibration uses seed 20260813. These settings document the frozen runs, without asserting identical results across arbitrary systems.

The reporting layer reads frozen simulation, calibration, and diagnostic outputs without re-estimating models. Reporting consistency checks and source/output hash manifests connect those records to the numerical tables. Figure-data checks verify all 480 plotted values against the frozen reporting CSVs, while the figure manifest records source and final-image hashes. Full supplementary tables, RMSE figures, detailed diagnostics, configuration files, and environment records remain in the repository even where they are not printed in the current PDF.

Declaration

I hereby affirm that I have written the above thesis independently and have used no sources or aids other than those indicated; that the submitted work has not previously been submitted for examination at any other university; and that it has not been published, either in whole or in part. I have clearly indicated all direct quotations and passages paraphrased from other works as such.

September 17, 2026

Olimjon Umurzokov